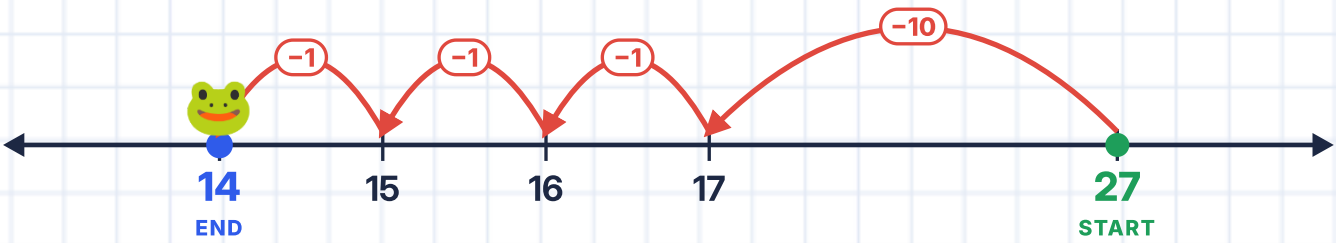
START HERE

Big Jumps on a Number Line

How to add and subtract **big numbers** like **27 − 13** on a number line, step by step, with **Hoppy the frog** 🐸.



IN THIS BOOK YOU WILL LEARN

1. The 3 questions
2. The empty number line
3. Where to write the start
4. Breaking a number into tens and ones
5. The jump-of-10 trick
6. 27 − 20: one jump or two?
7. 27 − 13 step by step
8. Adding: 34 + 25
9. The recipe
10. Oops! Common mistakes
11. Practice + answers

My name: _____ **Class:** _____

The 3 questions

For every number line sum, ask these **3 questions**, in this order:

1. **Where do I START?**
   Always on the **FIRST** number.

2. **Which WAY do I jump?**
   Look at the sign. **Plus + → forward (right)**. **Minus – ← backward (left)**.

3. **How BIG are my jumps?**
   Look at the **SECOND** number. Break it into **tens** and **ones**.

EXAMPLE: 27 – 13 **27** **–** **13** **=** **?**

1. Start on **27** 2. Minus → **backward** 3. 13 = **1 ten and 3 ones**

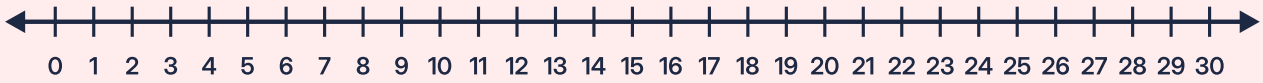
REMEMBER

START? WAY? HOW BIG?

The empty number line

For big numbers we do **not** write every number. That is too slow and messy!

× TOO MANY NUMBERS



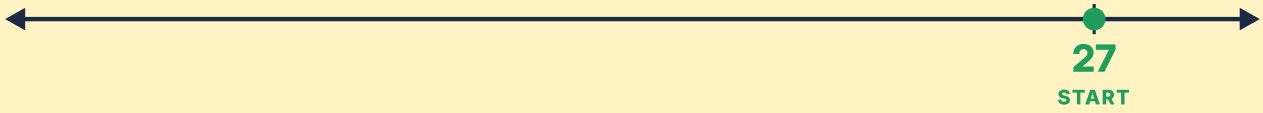
✓ THE EMPTY (OPEN) NUMBER LINE

Draw just a line. Write **only** the numbers you start on and land on.

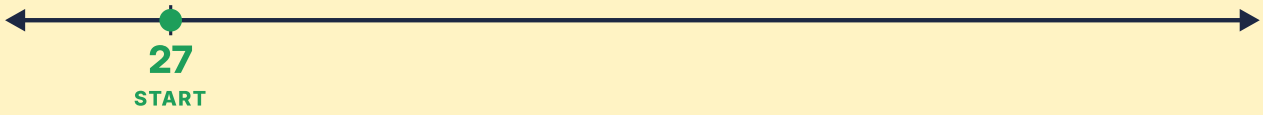


WHERE DO I WRITE THE START NUMBER?

Minus: Hoppy will go **left**. So write the start near the **RIGHT end**, to leave room on the left.



Plus: Hoppy will go **right**. So write the start near the **LEFT end**, to leave room on the right.



How big are my jumps?

Look at the **second number**. Break it into **tens** and **ones**, just like place value.

1. **13 = 1 ten + 3 ones**
   → **1 jump of 10**, then **3 jumps of 1**.

2. **20 = 2 tens + 0 ones**
   → **2 jumps of 10**, and **no** small jumps.

3. **25 = 2 tens + 5 ones**
   → 2 jumps of 10, then 5 jumps of 1.

REMEMBER

The **tens digit** tells how many jumps of 10. The **ones digit** tells how many jumps of 1.

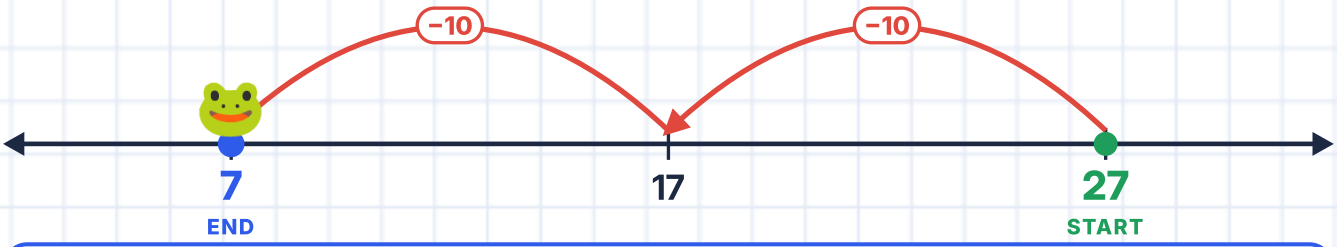
YOUR TURN

Number	Jumps of 10	Jumps of 1
31	_____	_____
14	_____	_____
40	_____	_____
8	_____	_____
52	_____	_____

The jump-of-10 trick

When Hoppy jumps **back 10**, only the **tens digit** goes down by 1. The **ones digit stays the same!**

27 → 17 → 07 = 7



JUMPING FORWARD 10 WORKS THE SAME WAY

34 → 44 → 54

Only the tens digit goes **up** by 1.

DO NOT DO THIS

Do not count back 10 small steps one by one (26, 25, 24 ...). It is slow and easy to make a mistake. Just change the tens digit!

YOUR TURN

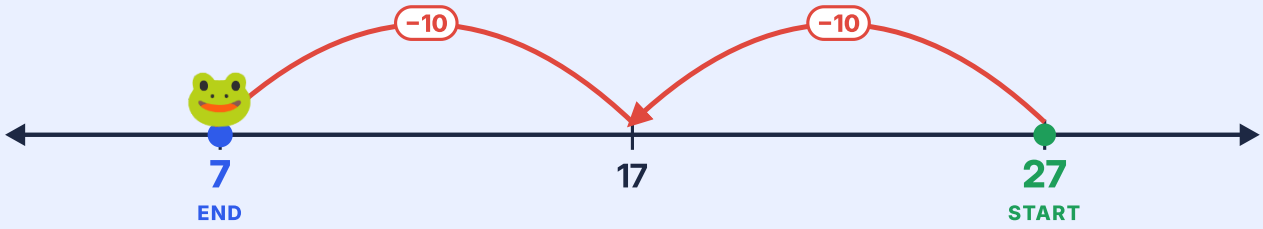
10 less than 45 = _____ 10 less than 63 = _____ 10 more than 34 = _____

10 more than 58 = _____

27 – 20: one jump or two?

20 is **2 tens**. Should Hoppy make **two jumps of 10** or **one jump of 20**? Let us try both!

WAY 1: TWO JUMPS OF 10



27 → 17 → **7**

WAY 2: ONE BIG JUMP OF 20



27 → **7**

BOTH WAYS ARE CORRECT!

Hoppy lands on the **same number, 7**. One jump of 20 is just two jumps of 10 **joined together**.

2 jumps of 10 = 1 jump of 20

WHICH ONE SHOULD I USE?

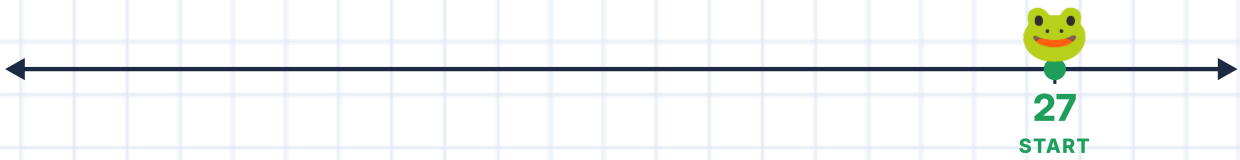
While you are learning: use jumps of 10, one ten at a time. It is easier to count and check.

When you are very sure, you can join them into one big jump.

27 – 13, step by step

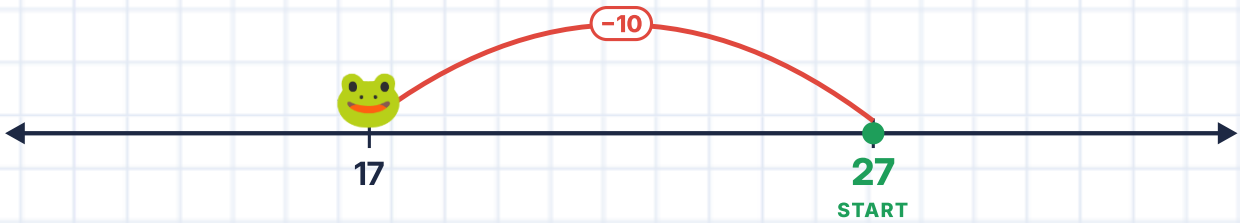
27 – **13** = **?**

1. START on the first number, 27. Minus → backward, so write 27 near the right end.

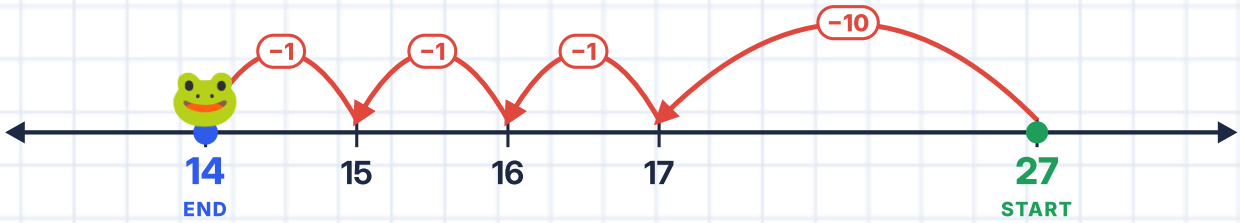


2. Break 13: **13 = 1 ten + 3 ones**

3. Tens first! Jump back 10. Only the tens digit changes: 27 → 17.



4. Now the 3 ones: jump back 1, three times: 16, 15, 14.



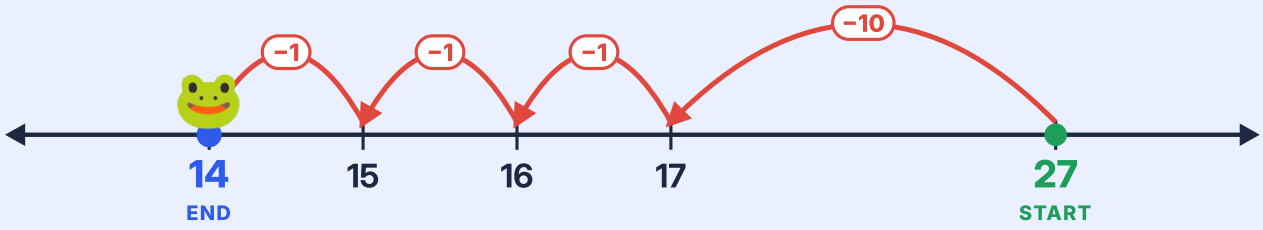
5. Hoppy landed on 14. That is the answer.

27 – 13 = 14 Check: 14 + 13 = 27 ✔

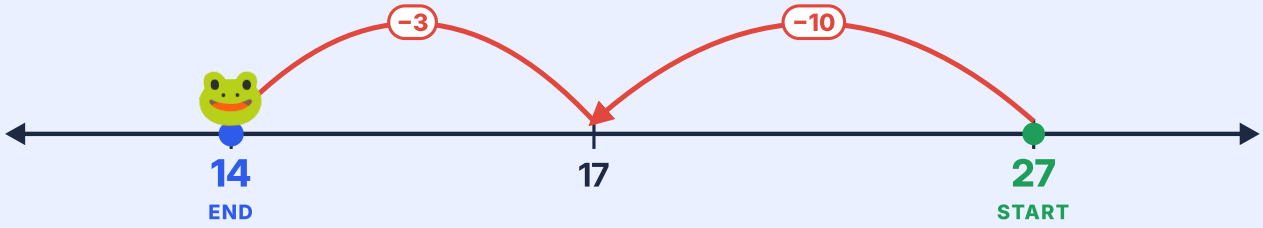
27 – 13: the fast way

Instead of three small jumps of 1, make **one jump of 3**. Same answer!

LEARNING WAY: 10, THEN 1, 1, 1



FAST WAY: 10, THEN 3

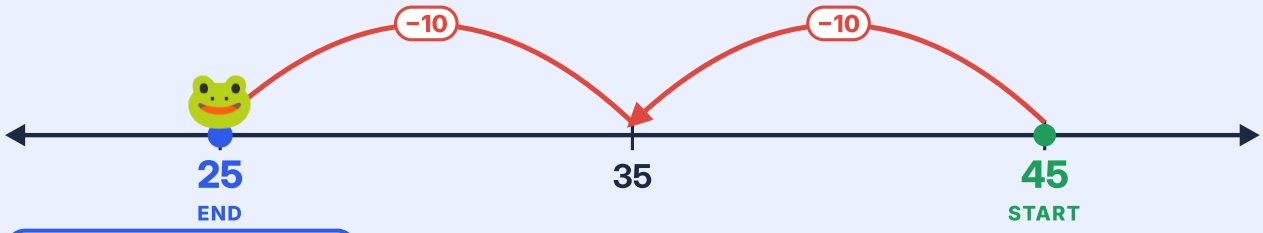


REMEMBER

Small jumps or one bigger jump, Hoppy lands on the **same place**. Always do the **tens first**, then the ones.

ONE MORE: 45 – 20

Start 45 · backward · 20 = 2 tens, 0 ones → 35, 25.

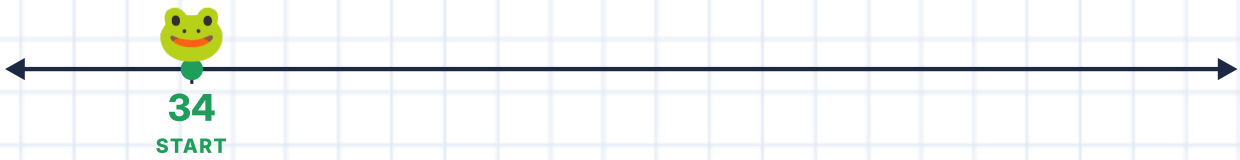


45 – 20 = 25

Adding: 34 + 25

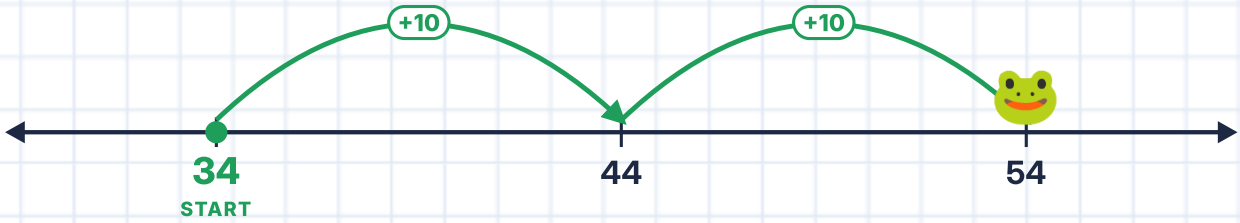
34 + **25** = **?**

1. START on 34. Plus → forward, so write 34 near the LEFT end.

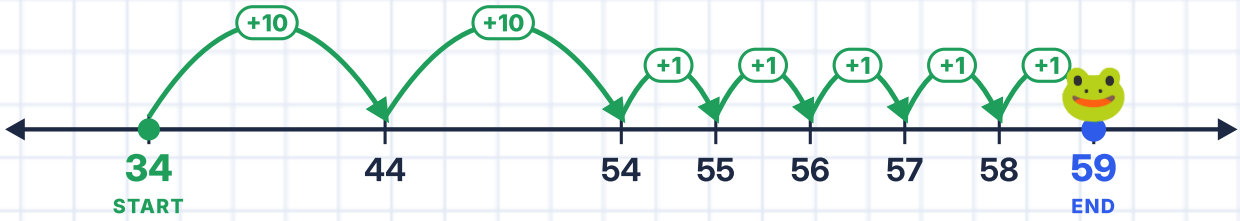


2. Break 25: **25 = 2 tens + 5 ones**

3. Tens first: jump forward 10, two times: 44, 54.



4. Now 5 ones: 55, 56, 57, 58, 59. (Or one jump of +5.)



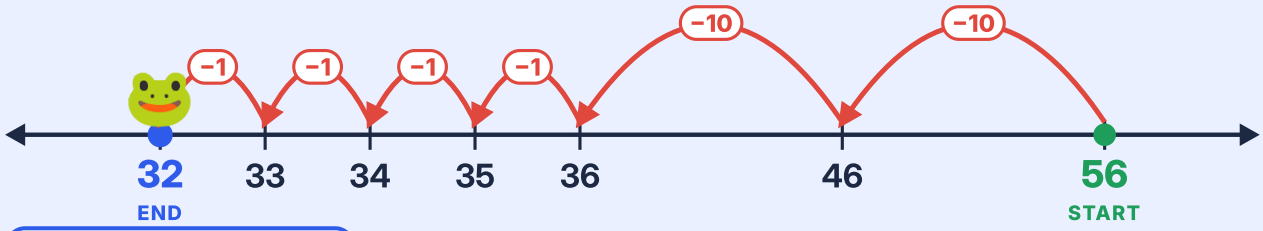
5. Hoppy landed on 59.

34 + 25 = 59

More examples

56 − 24

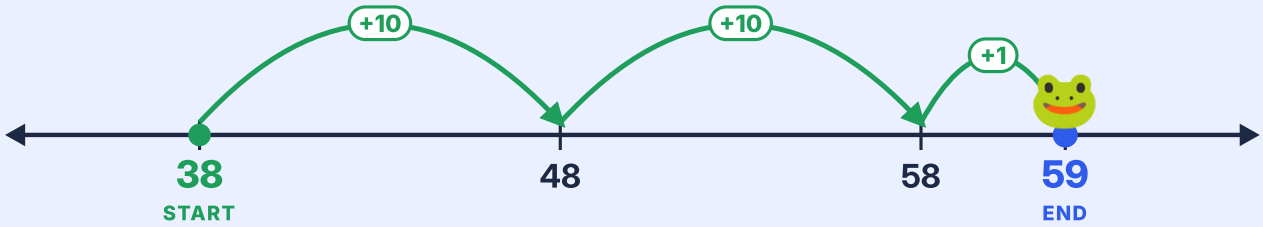
Start **56** · **backward** · 24 = **2 tens + 4 ones** → 46, 36, then 35, 34, 33, 32.



56 − 24 = 32

38 + 21

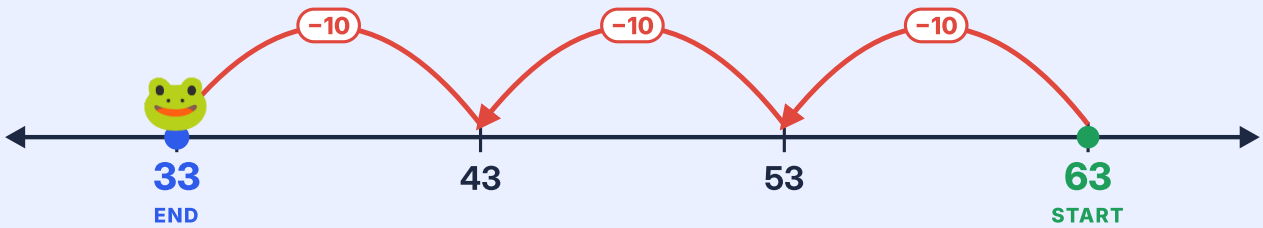
Start **38** · **forward** · 21 = **2 tens + 1 one** → 48, 58, then 59.



38 + 21 = 59

63 − 30

Start **63** · **backward** · 30 = **3 tens + 0 ones** → 53, 43, 33.



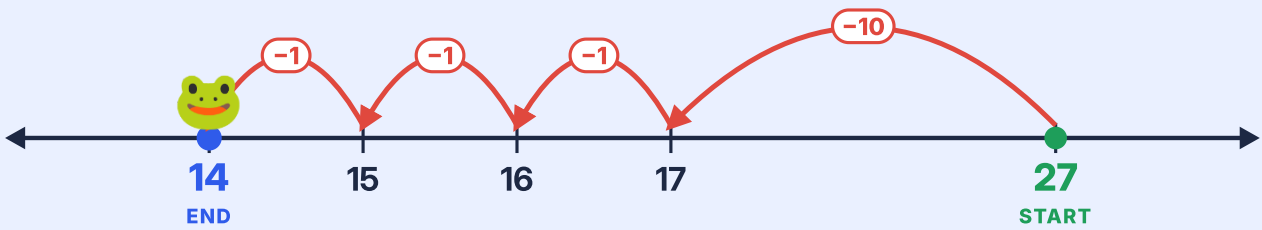
63 − 30 = 33

REMEMBER

The number line recipe

1. **Start on the FIRST number.**
2. **Look at the sign.**
   − → backward, write the start near the right end. + → forward, write the start near the left end.
3. **Break the SECOND number into tens and ones.**
4. **Jump the tens first, 10 at a time.**
   Only the tens digit changes.
5. **Then jump the ones.**
6. **Where Hoppy lands is the answer. Check it with the opposite sum.**

ALL IN ONE PICTURE: 27 − 13



WATCH OUT

Oops! Common mistakes

MISTAKE 1: STARTING ON THE SECOND NUMBER

27 − 13 does **not** start on 13. Always start on the **first** number: 27.

MISTAKE 2: GOING THE WRONG WAY

Minus goes **backward** (left), so the answer must be **smaller** than the start. If your answer for 27 − 13 is bigger than 27, you went the wrong way.

MISTAKE 3: WRONG NUMBER OF TENS JUMPS

13 has only **1** ten → one jump of 10. 20 has **2** tens → two jumps of 10 (or one jump of 20).

MISTAKE 4: COUNTING 10 SMALL STEPS

A jump of 10 is **one** jump. Just change the tens digit: 27 → 17.

MISTAKE 5: COUNTING THE START

Hoppy sitting on 27 is **not** a jump. Count only when Hoppy **lands**.

CHECK YOURSELF

Do the opposite sum. 27 − 13 = 14, so 14 + 13 should be 27. ✔

PRACTICE

Your turn! (1)

Use the empty number line. Write the start, draw the jumps, write each number Hoppy lands on.

1. 27 − 13 = _____



Start on _____ Way: forward / backward _____ tens and _____ ones
Answer: _____

2. 45 − 20 = _____



Start on _____ Way: forward / backward _____ tens and _____ ones
Answer: _____

3. 38 + 21 = _____



Start on _____ Way: forward / backward _____ tens and _____ ones
Answer: _____

PRACTICE

Your turn! (2)

4. 56 − 24 = _____



Start on _____ Way: forward / backward _____ tens and _____ ones
Answer: _____

5. 63 − 30 = _____



Start on _____ Way: forward / backward _____ tens and _____ ones
Answer: _____

6. 49 − 15 = _____



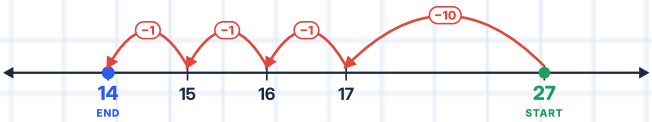
Start on _____ Way: forward / backward _____ tens and _____ ones
Answer: _____

Answers

Step 3: 31 → 3 and 1 · 14 → 1 and 4 · 40 → 4 and 0 · 8 → 0 and 8 · 52 → 5 and 2

Step 4: 35 · 53 · 44 · 68

1. 27 − 13 = 14



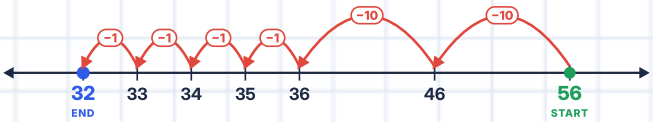
2. 45 − 20 = 25



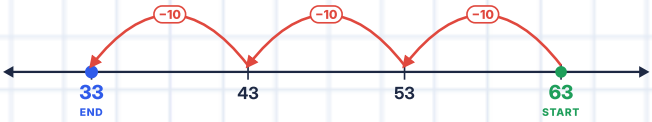
3. 38 + 21 = 59



4. 56 − 24 = 32



5. 63 − 30 = 33



6. 49 − 15 = 34

